LEARNING OUTCOMES!

Candidates should be able to:

- describe, with the help of a simple diagram for each method, how non-renewable fuels (coal, crude oil and natural gas) are extracted: – coal as obtained by open cast, adit and shaft mining methods – natural gas and crude oil obtained by exploration and drilling
- understand the difference between renewable and non-renewable sources of electricity
- explain (briefly) how electricity can be generated from renewable resources (hydel, wind, solar, and other possibilities such as wave, tidal, biofuels, geothermal)
- understand the importance of power sources for development.
- describe the quality and the amount of coal available from within Pakistan and how long reserves are likely to last, and also describe the types of coal which have to be imported for industrial purposes
- describe how coal both produced in Pakistan and imported is transported to the end users
- state how much natural gas is produced by Pakistan, and how long reserves are likely to last
- describe the extent of the natural gas pipeline network in Pakistan and explain how natural gas can be taken to those parts of Pakistan away from the pipelines, and the limitations of doing this
- state how much oil is produced by Pakistan, how long reserves will last and how much oil is imported, and explain why it is necessary to import large amounts of oil
- describe the extent of the oil pipeline network in Pakistan and describe the other methods that are used to transport both imported oil and oil produced in Pakistan
- understand that electricity can be generated in a variety of ways. In thermal power stations by burning coal, oil, gas and waste, or with nuclear energy; or with renewable sources e.g. water (including hydel), the wind and the sun
- understand that non-renewable power sources are running out, and are increasing in price.

- explain and evaluate the advantages and disadvantages of the different methods of producing electricity from renewable resources (generated by water, wind, wave and sun)
- understand the physical and human conditions that favour the development of multi-purpose hydel schemes
- state and explain the factors, both physical and human, which promote or hinder the availability of electricity and other power resources listed, including the feasibility of small-scale, renewable power generation
- explain why the supply of electricity is not sufficient or reliable to develop many parts of Pakistan.

COAL

- ✓ Coal is basically formed by the decomposition of plants.
- ✓ Plants are buried in the subsurface and when they decompose they form coal.
- ✓ Coal is always found in layers.
- ✓ Hence forms of coal are:
- ✓ Peat → Lignite → Sub-Bituminous → Bituminous → Anthracite

DETERMING QUALITY OF COAL

- ✓ It is determined by at what depth it is found.
- ✓ Quality of coal depends upon its color. Darker the colour, better the quality.
- ✓ It also depends upon its carbon content. The more carbon the better the coal is.
- ✓ Depends upon its heating value. The more heating value the better the coal is.
- ✓ If there will be more sulphur, Coal is of inferior quality.
- ✓ If there will be less sulphur, Coal is of superior quality.
- ✓ More moisture, Coal is of inferior quality.
- ✓ Less moisture, Coal is of superior quality.

TYPES OF COAL

Anthracite

- ✓ Blackest coal of all and found in thin layers, deep underground.
- ✓ It is the best quality of coal, hardest, with highest carbon content and burns with great heat.

Bituminous

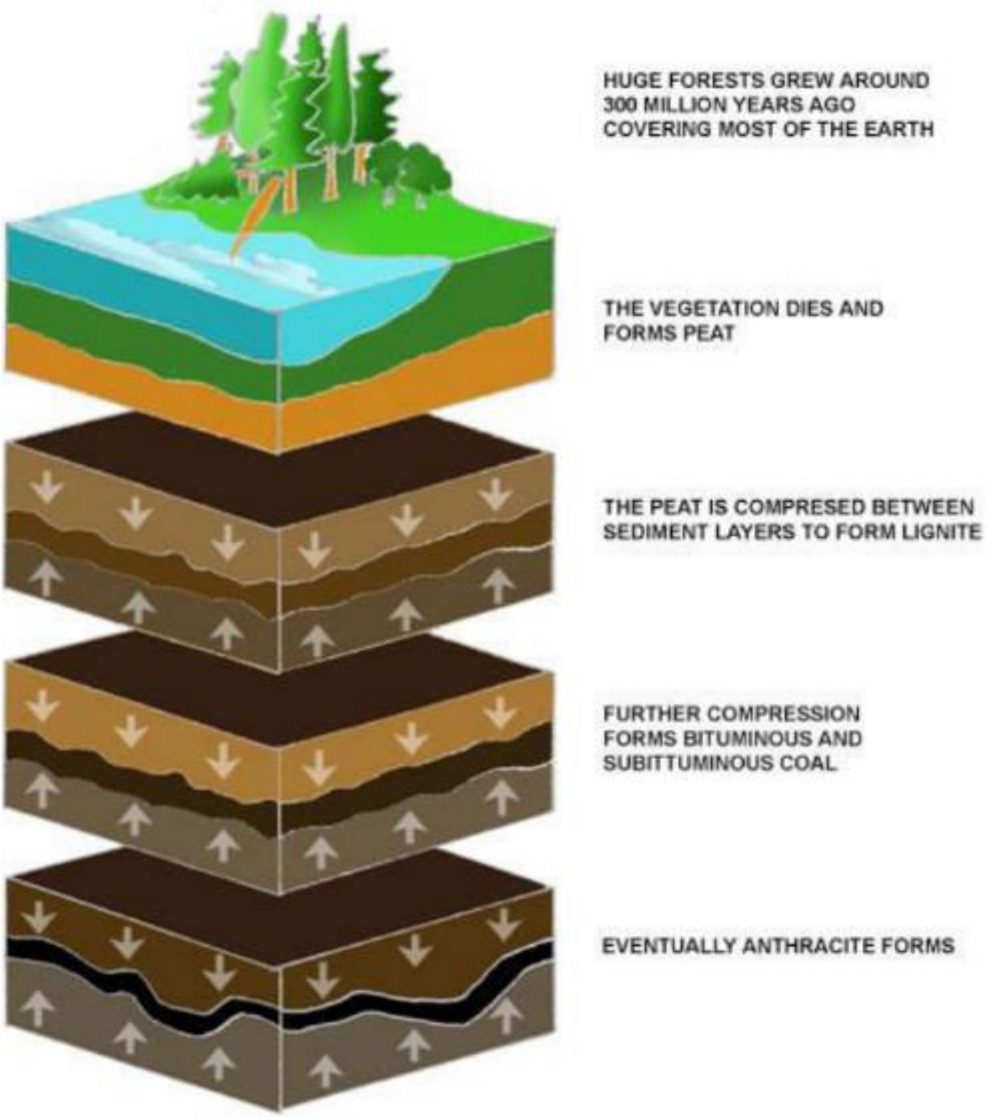
- ✓ It is darker in color and found further" deep underground and has two types; Steam Coal and Coking Coal

Lignite

- ✓ It is found near the surface and easier to mine. It is a low quality coal with a high moisture and ash content. It has a low heating value.

Peat

- ✓ It is exclusively vegetative matter and represent the initial stage of coal formation.



COAL IN PAKISTAN

- ✓ In Pakistan, lignite and sub bituminous quality of coal is found.
- ✓ Having: Low heating value, Low carbon content, High moisture and High sulphur content
- ✓ Coal region is the same as coal fields.
- ✓ Mines is the same as mining centers.
- ✓ So many mines combine to form a coal region.
- ✓ If coal is located near the surface then it is mined by Open-cast mining.

- ✓ If coal is located in the greater depth then it is extracted by Shaft mining.
- ✓ If coal is located on hill slopes then Adit mining is used to extract coal.

COAL EXTRACTION

The extraction of coal occurs by 3 main methods, which are Adit, Shaft and Opencast.

Adit Mining

- ✓ Adit method of mining is used when a mineral layer is exposed near the surface of a hill.
- ✓ In this technique a single near horizontal or multiple layers are dug into the mineral layer.
- ✓ Explosives are used to blast and loosen the rock.
- ✓ Once this has been carried out, pillars and steel nets are installed.
- ✓ This prevents the roof of mine from caving in and steel nets prevent rocks from falling down and killing the miners.
- ✓ Then diggers are used to remove the mineral bearing rock and which is then transported by rail or donkeys to the mine entrance after which is loaded into trucks

Shaft Mining

- ✓ Shaft mining is used when the mineral layer is found deep in the soil.
- ✓ First a vertical layer is dug to reach the mineral layer.
- ✓ Once the mineral layer is reached then a horizontal layer is dug in both sides into the mineral layer.
- ✓ Explosives are used to blast and loosen the rock.
- ✓ Once this has been carried out then pillars and steel nets are installed.
- ✓ This prevents the roof of mine from caving in and steel nets prevent rocks from falling down and killing the miners.
- ✓ Then diggers are used to remove the mineral containing rock after it has been blasted apart by dynamites.
- ✓ This is then transported by rail to the lift, whereas then it is lifted to the surface to be transported by trucks.
- ✓ It must be noted that ventilation shafts are also dug along the length of mine along with main shaft to prevent the build-up of explosive odourless gases like methane.

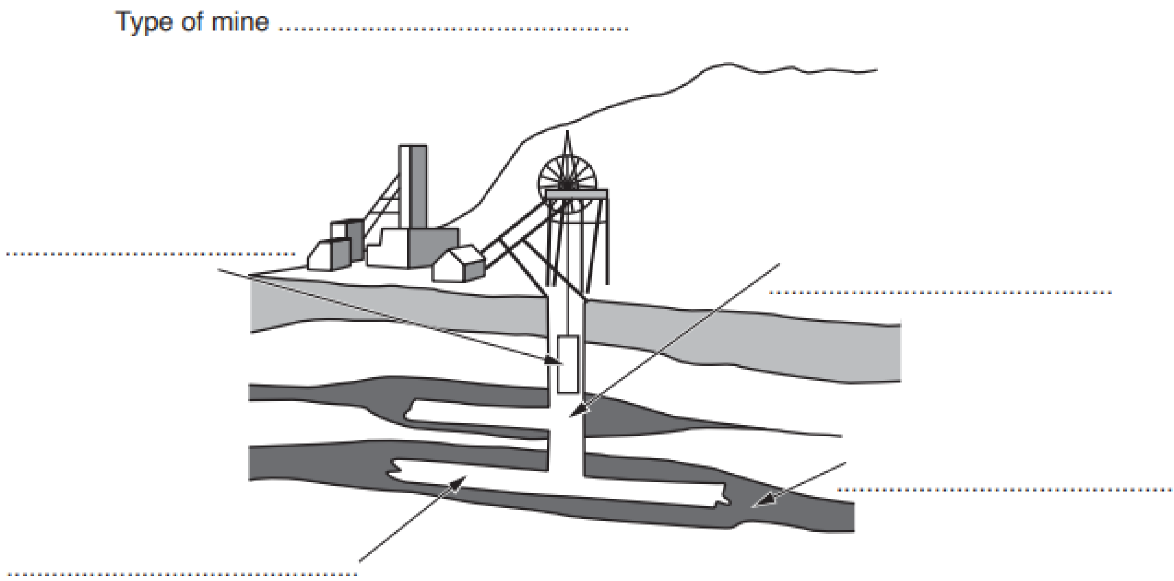
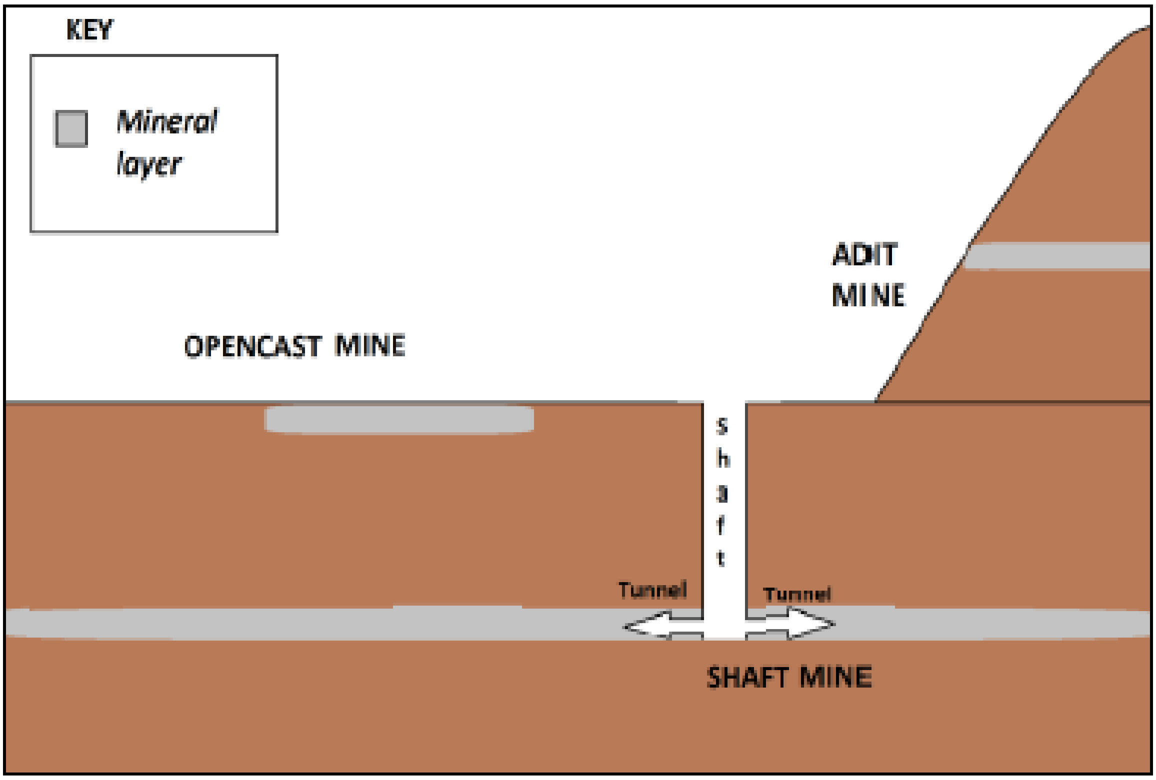


Fig. 4

- ✓ Type of Minining: Shaft Mining
- ✓ North Left: Cage
- ✓ South Left: Tunnel
- ✓ North Right: Shaft
- ✓ South Right: Seam

Open Cast Mining

- ✓ Opencast mining is used when the mineral layer is exposed near the surface of the earth.
- ✓ Firstly the vegetation is cut, topsoil and subsoil removed.
- ✓ Then explosives are used to blast and loosen the rock.
- ✓ Then diggers are used to remove the mineral containing rock which is then transported by huge trucks carrying 500 tonnes in one go onto the surface.
- ✓ Opencast mine is a big hole in the ground with pathways for trucks running on the diameter of the mine



THREE MAJOR COALFIELDS

- ✓ SALT RANGE (PUNJAB) and MAKARWAL-GULLAKHEL COALFIELDS (KPK) (LIGNITE TO SUB-BITUMINOUS)
- ✓ QUETTA COALFIELDS (BALOCHISTAN) (BITUMINOUS/SUB-BITUMINOUS)
- ✓ LOWER SINDH COALFIELDS (SINDH) (LIGNITE)

MINING CENTERS

Dandot

- ✓ Salt Range Coalfield.

Pidh

- ✓ Salt Range Coalfield.

Sor Range

- ✓ Quetta coalfield.
- ✓ 16 km east of Quetta.

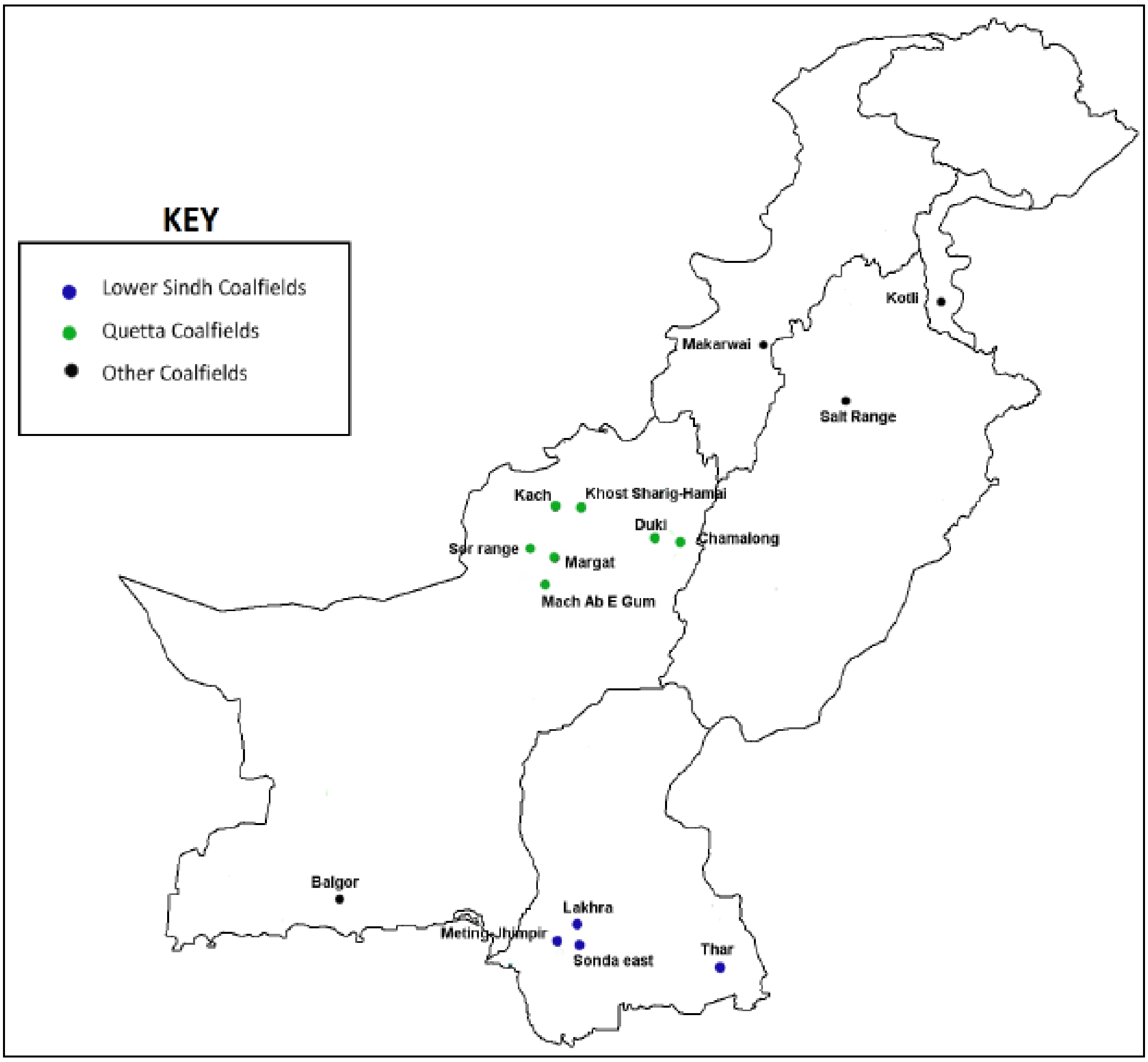
Mach

- ✓ Quetta coalfield.
- ✓ 55 km north of Quetta.

Lakhra

- ✓ Lower Sindh coalfield.
- ✓ North of Hyderabad.

LOCATION OF COALFIELDS



WHY PAKISTAN HAVE LOW QUALITY COAL?

- ✓ Lignite Coal.
- ✓ Low burning temperature.
- ✓ Low carbon content.
- ✓ High Ash content.
- ✓ High Sulphur content.
- ✓ Reserves not developed.
- ✓ Difficult to mine.
- ✓ Few industries use coal as a fuel.
- ✓ Mining for good quality is risky.

WHY COAL HAS TO BE IMPORTED?

- ✓ Poor quality coal.
- ✓ Lack of technology.
- ✓ Lack of capital.
- ✓ Lack of government interest.
- ✓ Lack of skilled labor.
- ✓ Anthracite and Bituminous is imported for Steel Industry.
- ✓ Mainly imported from India, Australia and Brazil.

USES OF COAL

- ✓ Brick Kiln
- ✓ Power Generation
- ✓ Steel Making (Imported coal is used)
- ✓ Industrial Uses
- ✓ Cooking

TRANSPORTATION OF COAL

- ✓ After the extraction of coal from the coal face, it is loaded onto trolleys, which run on a track, which leads from the coalmine to the outside surface.
- ✓ In some small coalmines donkeys are used as an underground transport.
- ✓ Once the coal comes out of the mine , the qualities of coal are separated and sold to the middleman who further loads it into trucks and supplies it to the brick kilns and cement factories where it used as a fuel.

- ✓ When the coal is supplied to thermal power stations, rail transport is also used if it is economically feasible.
- ✓ Brick kilns use 83 % of Pakistan's coal production.

CRUDE OIL

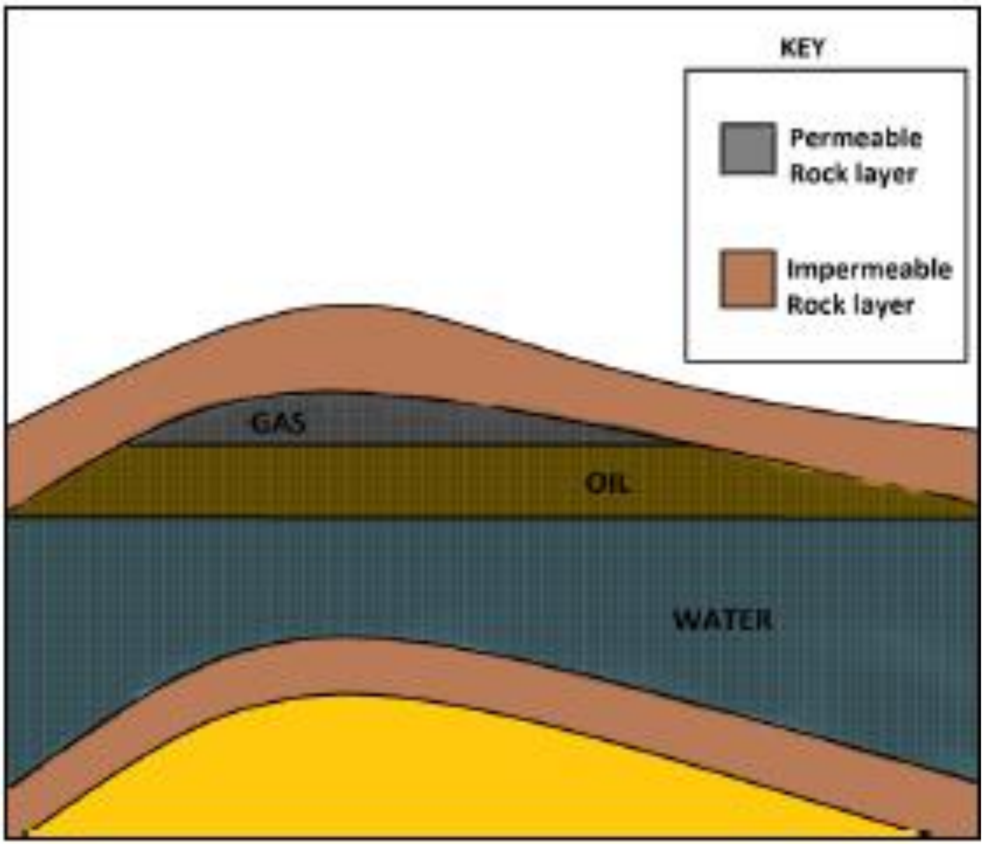
- ✓ Viscous and impure oil.

OIL DRILLING

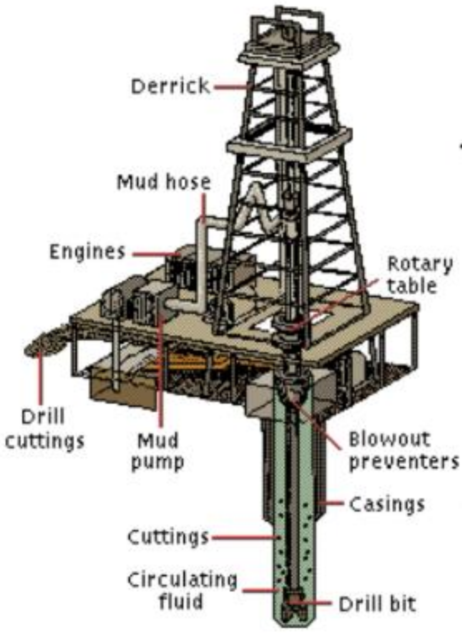
- ✓ Oil occurs in porous spaces of sedimentary rock and is derived mainly from decomposition of marine animals.
- ✓ Once drilling site has been selected, a derrick is setup.
- ✓ Derrick/drilling rig built
- ✓ Pipes inserted
- ✓ Diamond/tough metal drills into rock
- ✓ Oil rises when pressure released/pumped up
- ✓ Valves to control flow into pipeline
- ✓ Derrick removed/dismantled after oil is flowing

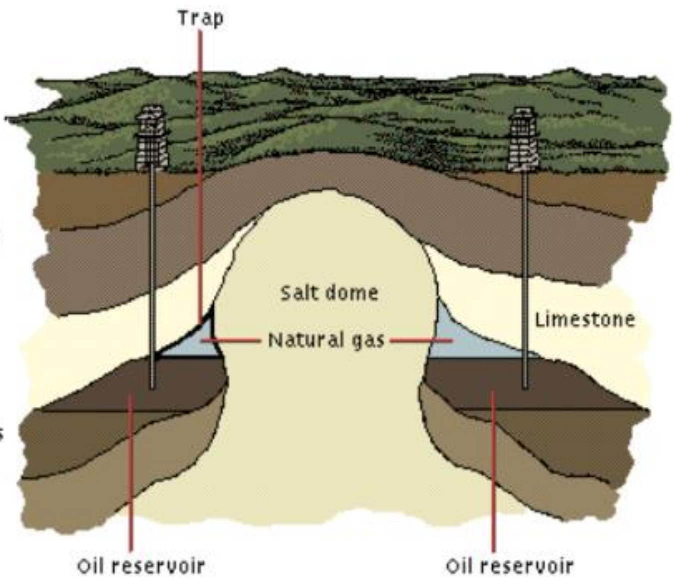
OIL TRAP

- ✓ Oil cannot get through rocks around it
- ✓ Oil trapped between layers of non-porous/impervious/impermeable rock
- ✓ Porous rock has pores to hold liquids/gases.



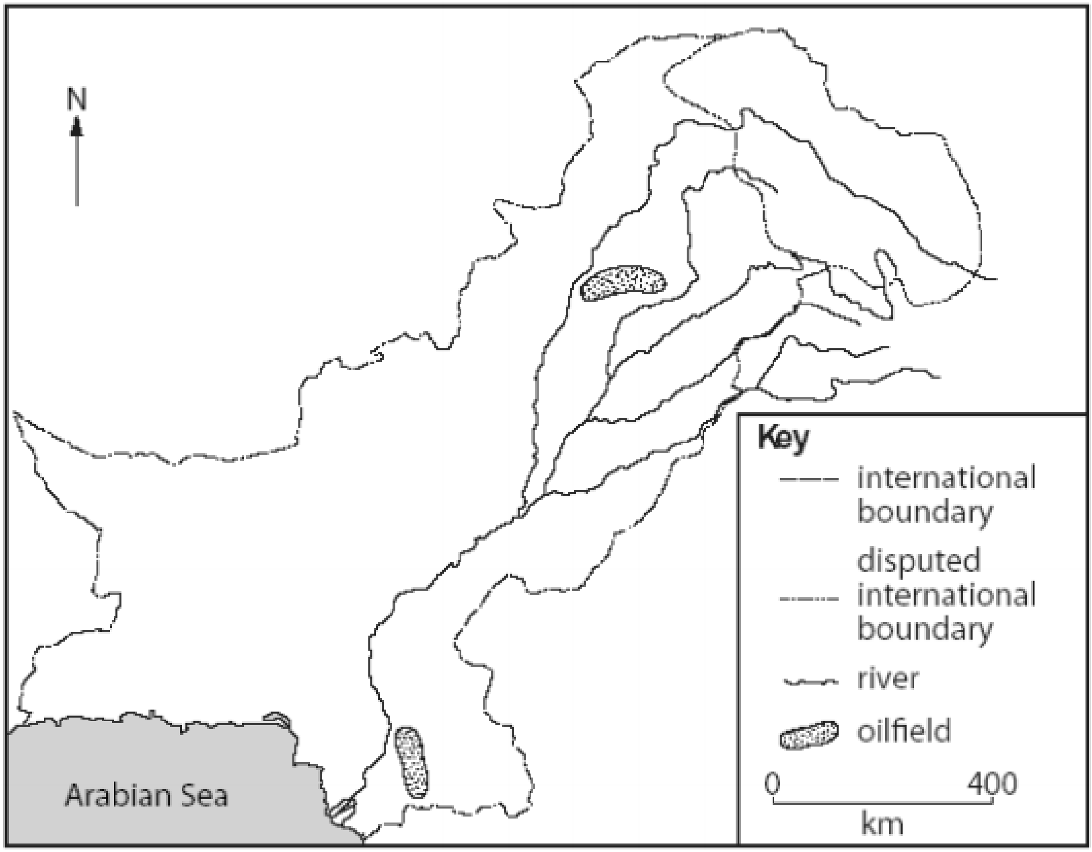
OIL DRILLING





OILFIELDS

- ✓ UPPER PUNJAB (Between River Indus & Jhelum)
- ✓ LOWER SINDH (Southern side) (Hyderabad)



WHY OIL REFINERIES?

- ✓ Crude Oil cannot be used in raw state.
- ✓ It has to be refined into useful products like Kerosene, Petroleum, Diesel and other Lubricant oil.
- ✓ These refineries process crude oil → through fractional distillation → after which different products are obtained e.g. wax.

OIL REFINERIES

- ✓ Attock Refinery – (Morga, Potwar Plateau)
- ✓ NRL – National Refinery Limited (Korangi, Karachi)
- ✓ PRL – Pakistan Refinery Limited (Korangi, Karachi)
- ✓ Parco Refinery – Pak Arab Cooperation (Multan, Mehmood Kot)
- ✓ Hub Refinery – (Hub, Baluchistan)

WHY DO WE IMPORT OIL?

- ✓ Pakistan produces 55,000 barrels per day.
- ✓ Requirement is 400, 000 barrels per day.
- ✓ Oil production is small
- ✓ Pakistan cannot satisfy its needs
- ✓ Resources not exploited
- ✓ Growing demand for population
- ✓ More industries
- ✓ Mechanization of agriculture
- ✓ Most thermal stations use oil

PROBLEMS OF IMPORTING OIL

- ✓ Negative trade balance
- ✓ Economy go down
- ✓ Uses for exchange
- ✓ Creditors increase influence over Pakistan's affairs
- ✓ Less money for investment and to spend in education, agriculture and health
- ✓ More tax is composed
- ✓ Cannot afford to exploit new oil fields.

TRANSPORTATION OF OIL

Imported Oil

- ✓ Imported Oil (from K.S.A, U.A.E., Kuwait)
- ✓ Sea imports by oil tankers at Kemari Port and Port Qasim
- ✓ Oil Pier (Platform with oil handling system)
- ✓ Oil products pumped from oil tankers
- ✓ Storage Tanks
- ✓ Pumped through Pipeline and Tankers
- ✓ Oil Refinery

Local Oil

- ✓ Transported by tankers from oil field → Refinery

Transportation of Oil on Land

- ✓ Tankers
- ✓ Rail
- ✓ Pipeline

Transportation of Oil from Kemari Port

- ✓ Pipeline
- ✓ Transported from Tanker
- ✓ From Oil Terminal at Kemari
- ✓ To NRL, PRL and Hub

Transportation of Refined Oil

- ✓ Road → Petrol Station
- ✓ Rail Tankers → Furnace oil → Thermal Power Station.
- ✓ Pipeline is also used

USES OF OIL

By Products

- ✓ Synthetics
- ✓ Plastics
- ✓ Pesticides

Transport

- ✓ Cars, trucks, railways etc.

Power

- ✓ Thermal power stations to produce electricity.

Industry

- ✓ Chemical, wax, rubber, plastic etc.

Domestic

- ✓ Detergents, heating, cooking.

Government

- ✓ Transport, public utilities.

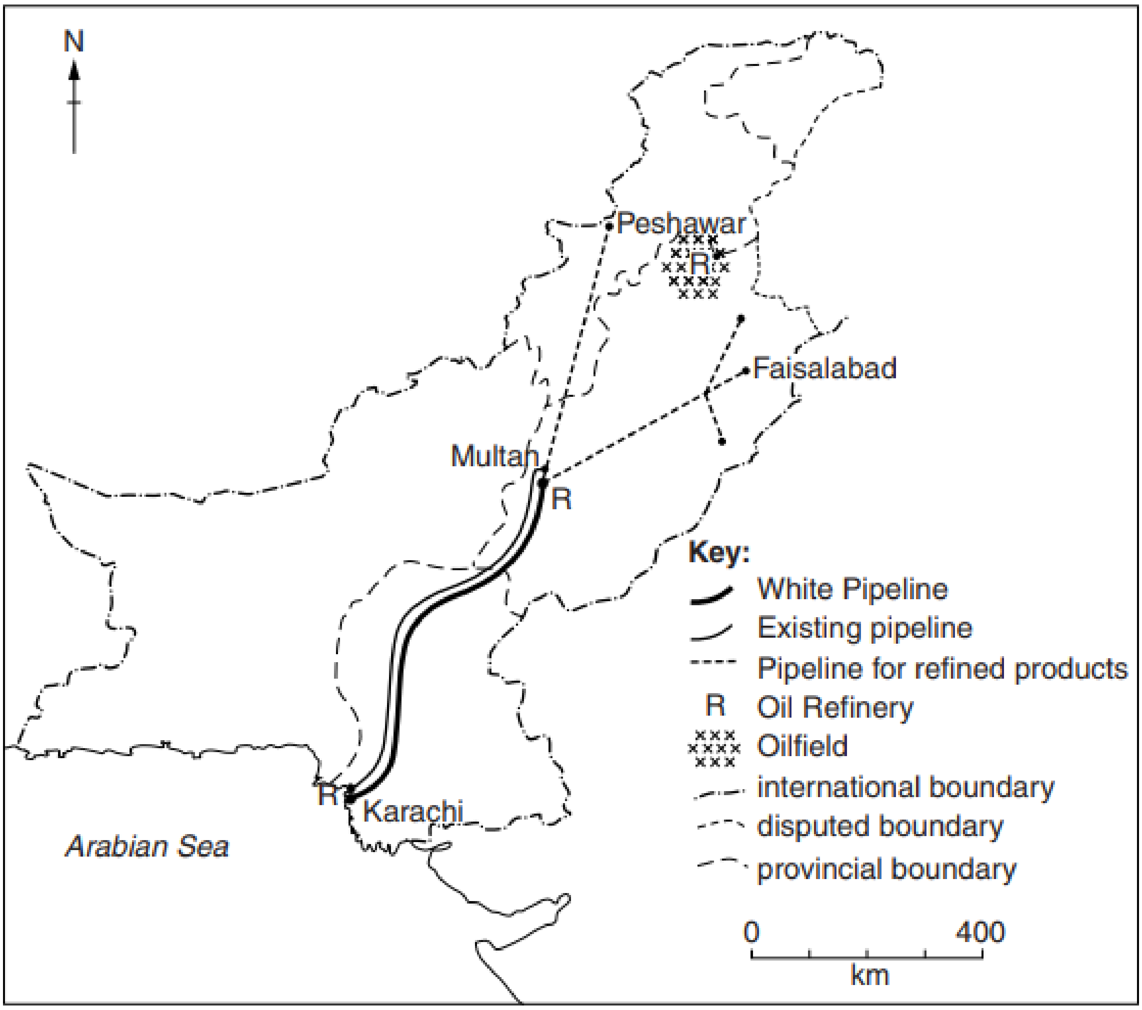
Agriculture

- ✓ Tube wells, insecticides, tractors, pesticides, for road making to practice agriculture.

WHITE OIL PIPELINE PROJECT

- ✓ In 2002 PARCO launched a white oil pipeline project (WOPP) which will carry refined oil from Karachi to the north.
- ✓ After conversions of PARCO's existing pipeline network for crude oil transportation, the white oil pipeline will be used for the transport of refined petroleum products to the central and northern regions of Pakistan.
- ✓ These areas account for almost 60 % of the total petroleum consumption in the country.
- ✓ Bin Qasim Port will be the initial point of the white Oil Pipeline project.

- ✓ The new underground pipeline costing $480 million will also carry refined oil from the Pakistan oil refinery at port Qasim to Mahmood Kot in district Muzaffargarh covering a distance of 817 KM.
- ✓ The demand for petroleum products is rising at a rate of 10 % per annum.



NATURAL GAS

- ✓ Made up of many gases methane, ethane, propane and butanes.
- ✓ When it is cooled to very low temperature it turns into a liquid (LPG).
- ✓ Colorless and Odourless

WHY SMELL ADDED TO NATURAL GAS?

- ✓ Colorless and Odourless.
- ✓ People will not be able to detect it until it becomes explosive.
- ✓ A chemical called Mercaptan is added to the gas.
- ✓ Mercaptan is harmless, non-toxic and has a strong "rotten egg" smell.

MAIN GAS FIELDS

- ✓ SUI (Balochistan) (7,423 million cubic metres)
- ✓ PIRKOH (Balochistan) (410 million cubic metres)
- ✓ MARI (Lower Sindh) (1876 million cubic metres)
- ✓ MEYAL (Potwar) (268 million cubic metres)
- ✓ DHURNAL (Potwar) (116 million cubic metres)

WHY NATURAL GAS EASY TO EXTRACT?

- ✓ small size of land
- ✓ little impact on the environment
- ✓ simple machinery/small machinery
- ✓ little investment needed
- ✓ pipes go into ground
- ✓ works automatically/no/little manpower needed

USES OF NATURAL GAS

- ✓ Power generation.
- ✓ Cooking and heating.
- ✓ For cement.
- ✓ For Fertilizers.
- ✓ For Vehicles.

SECTORAL CONSUMPTION OF GAS

- ✓ Power 29.44 %
- ✓ Fertilizer Industry 24.36 %

- ✓ Household 21.60 %
- ✓ Industry 19.40 %
- ✓ Commercial 03.27 %
- ✓ Cement Industry 01.89 %
- ✓ Transport 00.66 %

WHY NATURAL GAS CHEAP AND EASY TO USE?

- ✓ produced in Pakistan, not imported
- ✓ large reserves
- ✓ lightweight
- ✓ available in pipelines
- ✓ portable in cylinders
- ✓ cleaner than burning wood/coal
- ✓ easy to extract

TRANSPORTATION OF GAS

- ✓ Natural gas is transported on land by two means; pipelines and cylinders.
- ✓ The use of cylinders means that only little amount of gas can be carried at once.
- ✓ Refilling takes time, risk of explosion due to faulty cylinders is an added concern.
- ✓ The cylinders are heavy and difficult to move.

State two ways in which gas can be supplied to areas away from pipelines.

- ✓ Changed to a liquid/LPG/CNG
- ✓ Cylinders

WHAT CNG IS AND WHY IT IS USED?

- ✓ Compressed natural gas
- ✓ Methane under high pressure
- ✓ Compared to petrol / diesel cheaper
- ✓ Cleaner / reduces air pollution
- ✓ Safer
- ✓ Can be stored / transported in cylinders

GAS PIPELINE

LOAD SHEDDING

- ✓ Planned powercuts by the supplier of electricity is known as Load Shedding.

EFFECTS OF LOAD SHEDDING

- ✓ Interrupts production
- ✓ Damages machinery
- ✓ Cannot meet deadlines
- ✓ Loss of quality
- ✓ Loss of orders
- ✓ Loss of money/profit
- ✓ Cost of generators
- ✓ Lights/computers/freezers/air conditioning/heating etc. stops
- ✓ Transport/traffic problems

WHY ELECTRICITY CAN'T MEET THE DEMANDS?

- ✓ Population increasing so greater demand.
- ✓ Little new investment in new power stations (foreign investors less willing to invest due to political instability) (other government priorities such as healthcare/ education/housing/transport/alleviating poverty);
- ✓ Pakistan has small/inaccessible/depleting fossil fuel reserves (fossil fuels expensive to extract/poor quality/ have to import);
- ✓ Renewable energy plants expensive to construct;
- ✓ Power losses due to old/long transmission lines;
- ✓ Power theft (people diverting existing power sources for their own use);
- ✓ Most people live in rural areas (electricity does not reach there/lack of infrastructure/power lines);
- ✓ Many power plants are not working to full capacity (as a result of siltation in dams and reservoirs)/(they are still under construction);
- ✓ Power breaks down (lack of expertise to handle it)/(due to old machinery);
- ✓ More rural to urban migration (means demand cannot be fulfilled);
- ✓ Seasonal variations (less HEP generation in winter as less rainfall/snowmelt at times of peak demand).

MEASURES TO SOLVE ENERGY CRISIS

- ✓ Moving away from non-renewable to renewable
- ✓ Investment in large-scale power stations
- ✓ New Transmission lines to reduce power loss
- ✓ Reduce Theft by increasing security

- ✓ Energy saving in workplaces / homes
- ✓ Public / media awareness about not wasting energy resources

SUPPLYING ELECTRICITY TO RURAL AREAS

Possibilities

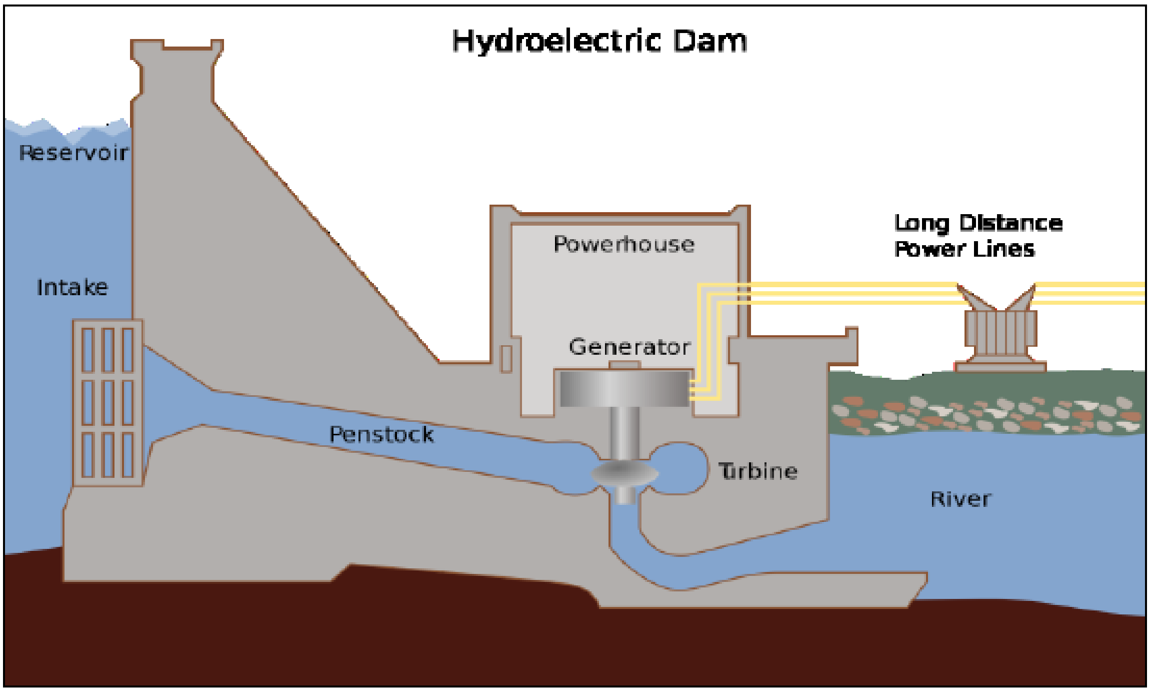
- ✓ Extend national grid
- ✓ Increase (national) power generation/nuclear power
- ✓ More/good potential for renewable schemes, wind, solar, HEP
- ✓ More small-scale power generation schemes E.g. biogas using animal/plant waste/molasses

Difficulties

- ✓ High cost of technology/fuel/maintenance
- ✓ Theft
- ✓ Damage/energy loss… …Due to long transmission lines/siltation in reservoirs for HEP
- ✓ Distance from grid stations/remoteness of some rural areas
- ✓ Tribal opposition
- ✓ Insufficient power generation… …So urban needs met first
- ✓ Lack of government support/loans/investment/policies
- ✓ Difficult construction in rugged/mountainous terrain
- ✓ Lack of skilled personnel, e.g. engineers

HYDROELECTRIC POWER

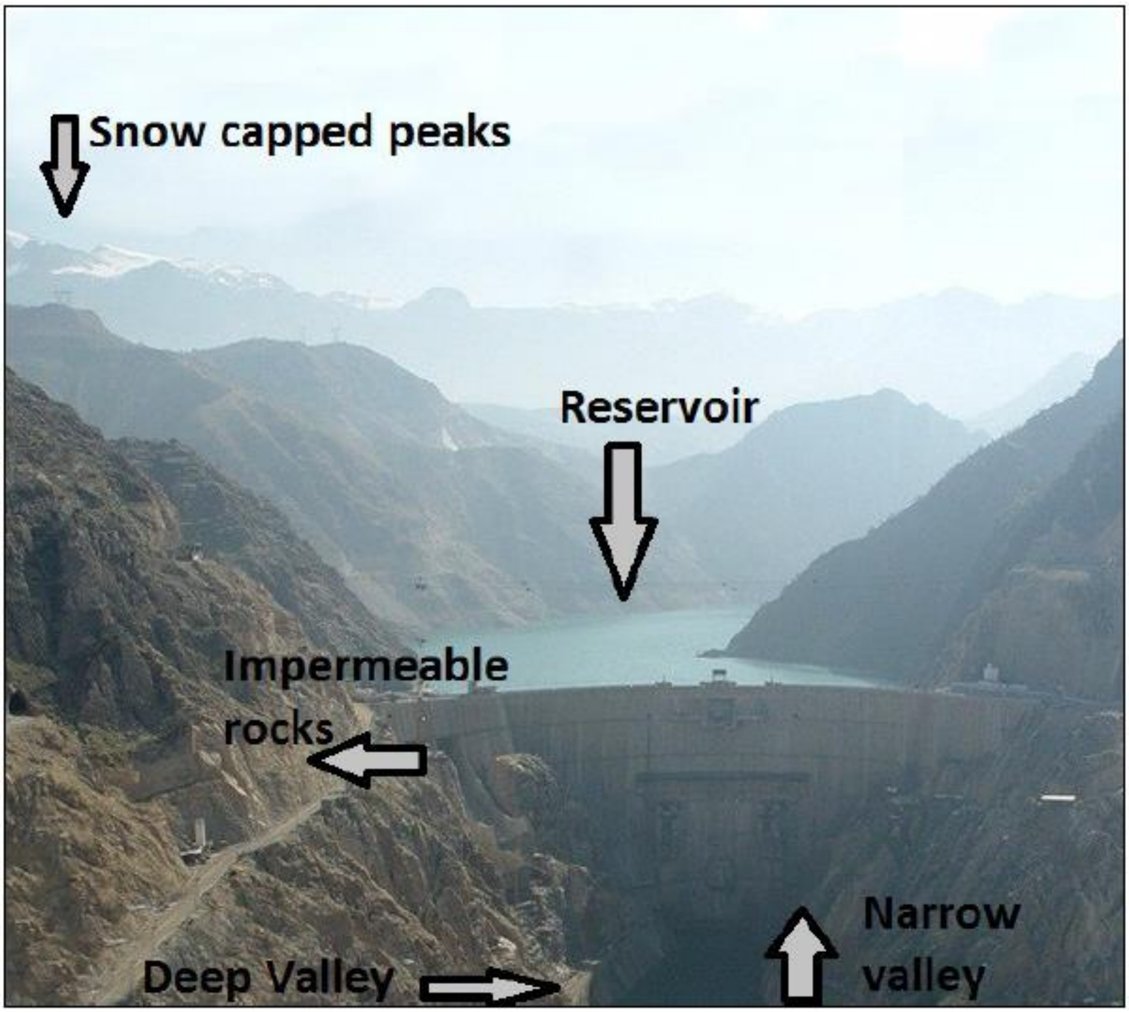
- ✓ This is the most important source of renewable energy in the world today.
- ✓ Hydroelectric power stations use the force of running water to turn turbines, which turn the generator in a magnetic field, thus generating electricity.



CONDITIONS THE LOCATION OF HEP SETUP

- ✓ A deep valley thus which can carry more water leading to more force at which turbines are turned. Also, a deep valley means that the water has more time to speed up and gather momentum before it hits the turbines.
- ✓ A narrow valley to make sure that construction cost of dam for concrete etc remain low and project is feasible due to the technical difficulties in building a wider dam wall.
- ✓ A large reservoir so to store any un-required surge of water which might have been wasted during rainy seasons. If this water is stored in the reservoir then it can be used in future months to generate electricity.
- ✓ The dam must not be located at or near a fault line. These areas are prone to earthquakes, which can fatally lead to dam failures, sudden flash floods and tragic loss of life.
- ✓ Impermeable layers of soil to prevent seepage and making sure that the reservoir of dam doesn't lose water too quickly.

- ✓ River basin that is fed by glaciers. The glaciers melt and provide water. Also, the drainage basin must be large and be under influence of the monsoons etc.
- ✓ River must pass through forested areas so that river carries low amount of silt. Trees roots bind the soil together and prevent flash floods (which erode more land) as they slow down the descent of rainfall. If this silt starts accumulating behind a dam wall it can cause some serious problems and reduce life of dams either by reducing their capacity or blocking spillways etc.
- ✓ The average temperatures of the surrounding areas should be low, to keep the rate of evapo-transpiration to a minimum.
- ✓ The dam must be located in a lowly populated area so the rehabilitation cost of citizens who are displaced is low.



Why it is difficult to develop more HEP (hydel) power stations in Pakistan?

- ✓ Climate change so less rainfall
- ✓ Climate change so higher temperatures and more evaporation

- ✓ Liable to siltation in reservoirs
- ✓ High cost
- ✓ No investment
- ✓ Opposition from tribal areas (in mountains)
- ✓ Security issues
- ✓ Lack of skilled labour

Why is HEP (hydel) an important source of electricity in northern Pakistan?

- ✓ Cheap to generate
- ✓ Renewable
- ✓ Available
- ✓ Rivers / water from glaciers
- ✓ High rainfall
- ✓ Lack of evaporation / lower temperatures
- ✓ Deep valleys for dams
- ✓ No air pollution

Why can the supply of power from these stations be unreliable?

- ✓ Shortage / not enough for every user
- ✓ Silting in reservoir (reduces capacity)
- ✓ Silt in turbines (causes damage)
- ✓ Seasonal shortages e.g. winter / frozen / monsoon etc.
- ✓ Lack of rainfall / changing climate
- ✓ Theft
- ✓ Damage to power lines

What problems occur when supplying electricity from reservoirs to areas of high population?

- ✓ Long distance to areas of use/high population
- ✓ Cost of wires and poles and Pakistan cannot afford due to shortage of money
- ✓ Loss by damage
- ✓ Loss by theft
- ✓ Loss of power by resistance/transmission

SOLAR

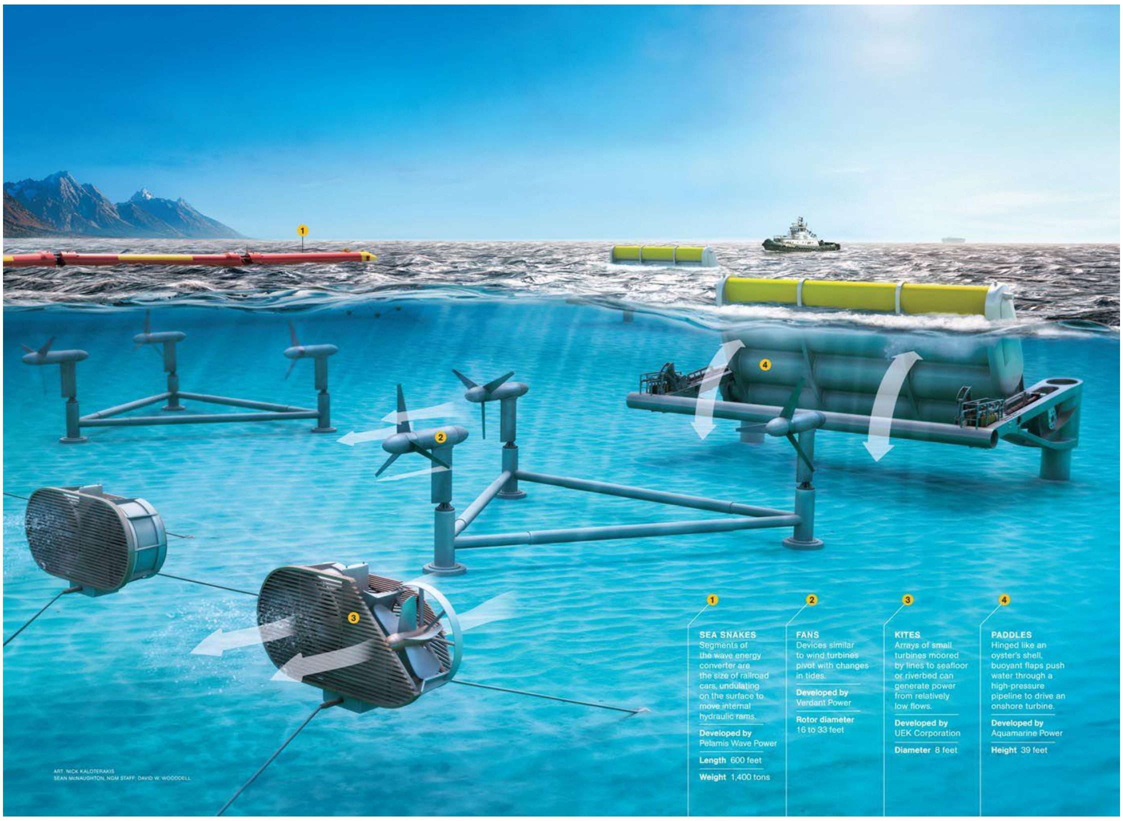
- ✓ Energy from the Sun can be utilized in the form of solar panels, which can generate electricity or heat water in homes.
- ✓ For generating electricity, these panels are installed in either deserts or large places of open and barren land.
- ✓ The panels absorb sunlight and use it to produce electricity free of cost except for the maintenance cost.
- ✓ These black panels absorb radiation from the sun and converts light energy into electrical energy.
- ✓ Another way of generating electricity is to use mirrors in desert, which reflect sunlight on to a very concentrated place on a tower (which contains water). The concentrated rays of sun heat the water and convert it into steam thus rotating the turbines and generating electricity.
- ✓ Always in south direction.

WIND

- ✓ Wind energy is based on simple principles.
- ✓ A turbine is attached to a blade.
- ✓ This blade is mounted on a tower which is placed in a windy area such as on hillsides, near cost or off-shore.
- ✓ The force of wind, turns blade and thus turns the turbine; producing electricity.

WAVES

- ✓ There is much need for research and development before we see any large scale use of this form of energy.
- ✓ Tidal power uses same principles as wind except being installed on seabed and using force of tides to turn the blades.
- ✓ Currently it is very expensive to mount these turbines on seabed and the electricity produced is low.



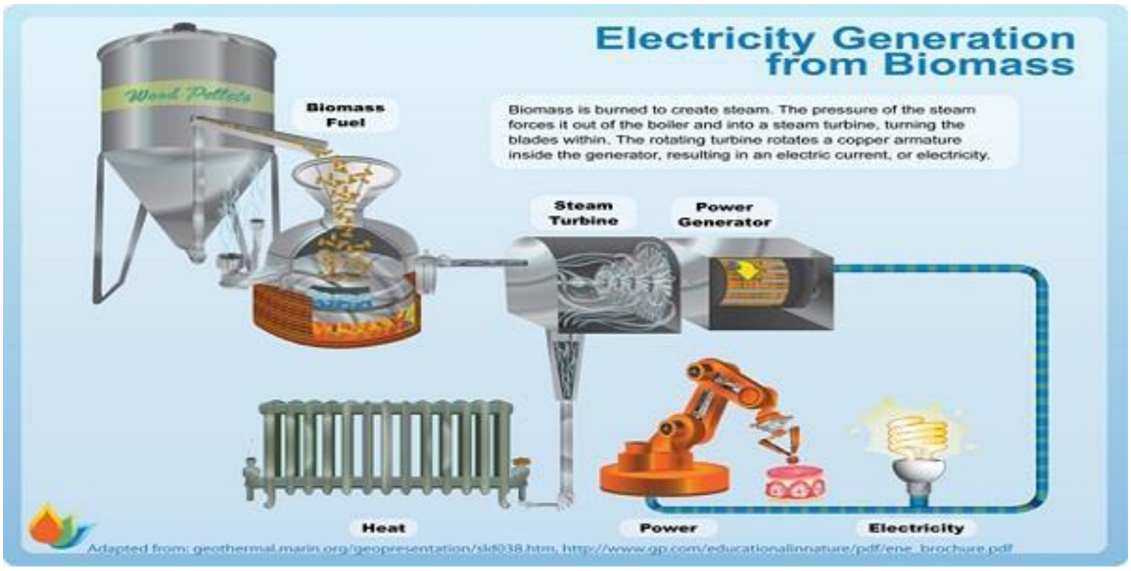
GEOTHERMAL

- ✓ Geothermal power is used in countries where favourable topographical conditions exist like in New Zealand and Iceland such as geysers etc.
- ✓ Conditions like these exist in some places in Azad Kashmir like near Kotli etc

- ✓ Two big pipes are dug into the surface.
- ✓ These pipes reach hot rocks in the earth's crust.
- ✓ These rocks are heated up by the magma beneath them.
- ✓ From one pipe cold water is sent below and from other pipe hot steam or water is received.
- ✓ Through a heat exchanger system this heat is transferred to water in another pipe network.
- ✓ Here the water absorbs heat and becomes steam which is fed into turbines to generate electricity

BIOMASS

- ✓ It is the energy generated in form of electricity or heat from plants/organic matter etc
- ✓ Dead leaves, branches, stumps etc are burnt directly to produce heat.
- ✓ In some countries garbage burning plants have been set up, which burn garbage and use the heat produced to turn water into steam.
- ✓ This steam then turns turbines, which produce electricity
- ✓ One good use of biomass is in production of biogas.
- ✓ In a biogas plant, animal dung is fed into a closed container deprived of oxygen.
- ✓ Bacteria and other micro-organisms then digest it and produce methane as a result.
- ✓ This methane is used either for cooking or for producing electricity



RENEWABLE RESOURCES

Definition

- ✓ Energy from a source that is not depleted when used, such as wind or solar power.

Advantages

- ✓ As their name suggests, they are renewable meaning that unlike non-renewable energy they will never run out.
- ✓ Renewable are mostly clean and have little lasting impact on environment.
- ✓ They require investment once only and maintenance thereafter, no need to buy fuel like fossil fuels etc. So they are cheaper in the long run.
- ✓ They can be used in remote areas (wind power in mountainous areas), where other power resources can't be used due to difficulty of transporting furnace oil etc. Also, these areas may be far away from the national grid (It is uneconomical to extend the grid to small population centres as the wires and pylons are expensive)
- ✓ The supply of fossil fuels is very unstable due to conflicts in the Middle East, thus to have secure energy supplies in the future, Pakistan should develop renewable energy.

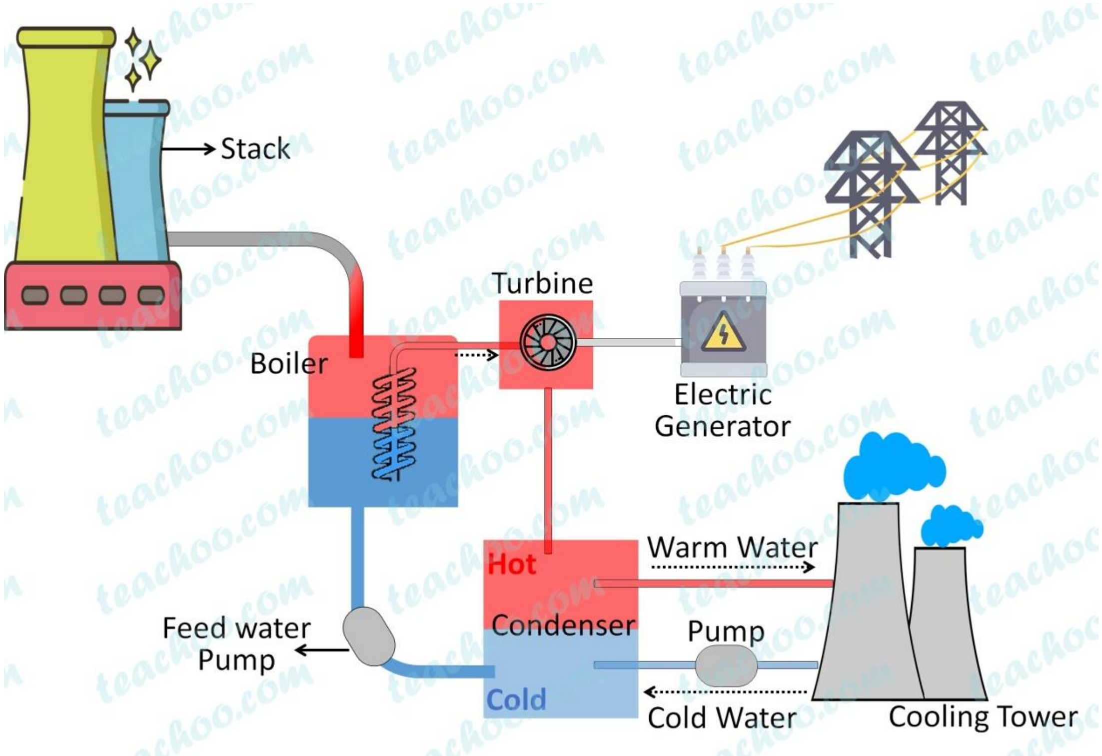
Disadvantages

- ✓ Solar energy is not much efficient, about 20%, when converting light energy from the sun into electricity; also it requires regular maintenance to remove dust which can cut efficiency by 3/4th. Also the panels must tilt with time of day and location of sun to maintain maximum efficiency
- ✓ Some dangerous substances like cadmium are used to make solar panels and raise questions about how they will be dealt in future when their lifetime has expired
- ✓ Wind turbines produce an audible sound which locals hate and it also spoils the scenery. Also not many places have many windy days
- ✓ Geo-thermal isn't available at all places in the world and few favourable places exist. Also it leads to Carbon Dioxide emissions which lead to global warming; melting of polar ice caps and floods of low lying areas
- ✓ Less water is available downstream due to dams, thus affecting fisheries as mangroves don't get enough fresh water. Agriculture may also be affected due to lack of water for plant growth
- ✓ Biomass produces limited amount of energy and is unfeasible to sustain current power demands

- ✓ Wind and solar power aren't available at all times like during windless days or cloudy days or nights
- ✓ All of the renewable sources require large amounts of capital to produce single megawatt of electricity as compared to fossil fuels
- ✓ In hydel projects immense costs are incurred due to; Large scale planning and research costs, which involves dam design etc, Evacuation procedures along with resettlement of people affected by the reservoir of the dam, Building roads (for access to site) and tunnels (to channel water into the turbines), Laying down power lines and connecting them to the grid system, hiring professionals (engineers etc) and workers and accommodating them on site, Buying expensive heavy machinery and maintaining it and Buying raw materials like cement and steel in exceptionally large quantities
- ✓ Hydro-electric causes massive displacement of people and loss of livelihood. Forests are flooded and killed; the trees then decay and produce methane, leading to global warming

THERMAL POWER

- ✓ Electricity that is generated by non renewable resources such as coal, oil and gas.
- ✓ Fossil fuels produce heat energy which is used to run water into steam which is then used to run turbines.



NUCLEAR POWER

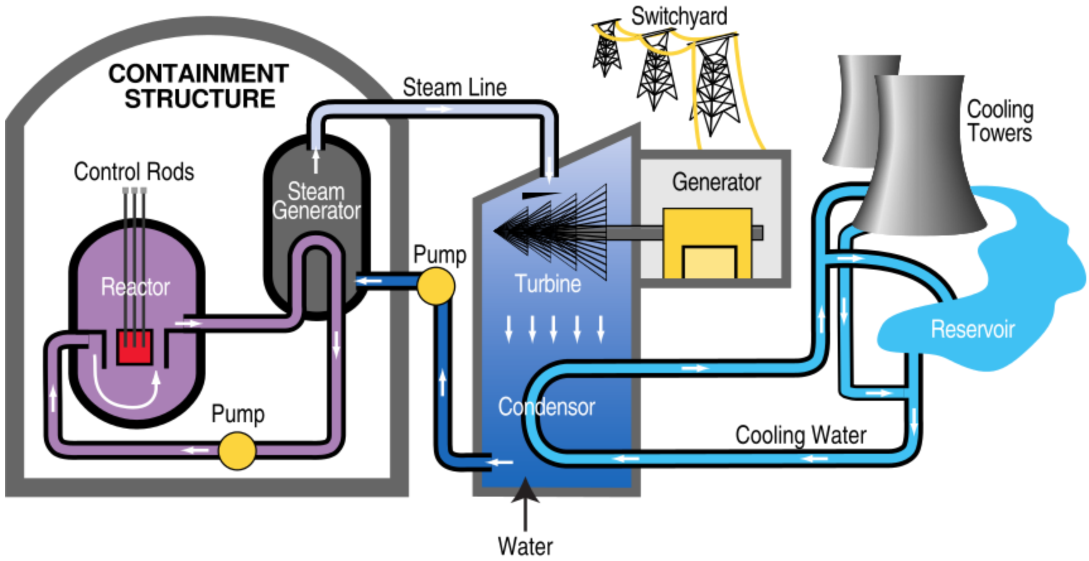
- ✓ Nuclear energy is power that is released from atoms and the most powerful source of energy.
- ✓ Fission: In atomic fission energy is released when atoms splits up into small substances.
- ✓ Fusion: In atomic fusion, energy is released when atoms are joined together to form a bigger atom that releases energy.
- ✓ Both process used heat energy to generate the electricity.
- ✓ Pakistan is also utilizing nuclear energy for electricity generation.
- ✓ Two nuclear power plants have been established in Pakistan; Karachi Nuclear Power Plant and Chashma Nuclear Power Plant.

Advantages

- ✓ large output
- ✓ reliable
- ✓ small input of raw material
- ✓ less pollution/environmentally friendly

Disadvantages

- ✓ expensive to buy fuel
- ✓ expensive to build
- ✓ lack of technology/skills
- ✓ dangerous/risk of radioactivity
- ✓ unpopular/local opposition
- ✓ risk of terrorism as used for bombs



NON RENEWABLE RESOURCES

Definition

- ✓ Non-renewable energy comes from sources that will run out or will not be replenished for thousands or even millions of years.

Advantages

- ✓ They require low initial investment as compared to renewable sources, and are perfect for urgently meeting demands as they require less time to build

- ✓ Electricity can be produced on large scale as no know how of advanced engineering like those involved in big dams etc is required
- ✓ The technology exists and therefore there are no research costs

Disadvantages

- ✓ These energy sources are non-renewable meaning that they will eventually run out and can't be depended upon in the long run
- ✓ They pollute the environment leading to acid rain or global warming. Acid rain kills trees and aquatic life. Also nuclear power produces wastes which remain radioactive for thousands of years and need to be stored underground safely for that period of time
- ✓ Due to conflicts in fossil fuel rich Middle East, the price of these fossils fuels looks set to rise, so Pakistan won't be able to import much oil to produce electricity in the future
- ✓ These projects however do require a lot of expensive maintenance and fuel, so are expensive in the long run. Also, **Independent Power Producers** charge a higher rate for electricity generation than state owned thermal stations
- ✓ Due to gas cuts in both industrial and power sector, it seems that relying heavily upon gas thermal stations isn't a reliable option in the long run

NATIONAL GRID SYSTEM

- ✓ The National Grid connects hydel generation in the north and thermal generation in mid country and the south managed by WAPDA and KESC.
- ✓ It consists of a large network of transmission lines and grid stations to transmit power to load centers and then to commercial and domestic consumers throughout the country.
- ✓ The purpose of forming a National Grid System is to supply electricity to different areas according to their requirements and not on the basis of their own power generation.
- ✓ For example some areas of Pakistan where heavy industries are located require more electricity than they are capable of generation supplying electricity through the National Grid System solves this problem.
- ✓ On the other hand, there are some areas where more electricity is generated than their actual requirements.
- ✓ This surplus electricity can then easily be transferred to other areas through this system.
- ✓ However, there is lot of wastage of electricity through the transmission lines and grid stations if they are not properly maintained, also due to the long distances involved.

SUSTAINIBILITY OF POWER

- ✓ Mismanagement of these resources can lead to energy crises for the future generation.
- ✓ In order to have sustainable development of power resources the following measures must be taken.
- ✓ The development of renewable resources of energy by using advanced technology.
- ✓ The preservation and conservation of the non-renewable resources of the earth to guard against the danger of their future exhaustion.
- ✓ The protection of the environment by enforcing the strict laws through environment protection.
- ✓ Faulty and damaged transmission lines should be replaced on an emergency basis to avoid losses in electricity.
- ✓ Strict measures to avoid the chances of theft.

6 MARKS

Question 1 **REVIEW BOOK BY MYM**

Read the following two views about solutions to Pakistan's shortage of energy:

A

To produce more energy, large-scale power generation schemes such as nuclear, gas-fired, oil-fired, and multi-purpose HEP (Hydel) power stations should be built.

B

To produce more energy, small-scale power generation schemes should be set up, such as biogas, wind, and solar power plants.

Which view do you agree with more? Give reasons to support your answer and refer to examples you have studied. You should consider both View A and View B in your answer. [6]

I agree with view B more.

Why B?

- ✓ Lower cost to maintain small scale schemes
- ✓ Renewable resources do not deplete
- ✓ Renewable resources do not pollute the environment
- ✓ Biogas – cheap source of energy
- ✓ Wind – available land in Balochistan highlands
- ✓ Solar – many parts of Pakistan experience 250–300 sunny days per year

Why not A?

- ✓ Large sums of money/loans needed
- ✓ Problems with disposing of waste
- ✓ Danger of flooding
- ✓ Coal extracted in Pakistan is poor quality for power generation
- ✓ Oil expensive to import
- ✓ Fossil fuel reserves are depleting
- ✓ Political issues between provinces with the construction of multipurpose dams over division of water

Question 2 REVIEW BOOK BY MYM

To what extent can Pakistan rely on fossil fuels to increase fuel and power supplies? [6]

Possibilities

- ✓ Large reserves of gas Sui / Pirkoh / Mari / Potwar Plateau area
- ✓ Large / new reserves of coal
- ✓ Coal a cheap fuel
- ✓ Potential of coal gas
- ✓ Port at Karachi for imports

Problems

- ✓ Small oil reserves / oil has to be imported
- ✓ Not renewable
- ✓ Coal is heavy / bulky to transport
- ✓ Gas is difficult to transport / explosive
- ✓ Fossil fuels expensive to import
- ✓ Higher cost of production / exploration / extraction

Question 3 REVIEW BOOK BY MYM

Pakistan is planning to expand its nuclear energy capacity from 1300 to 8800 megawatts between 2018 and 2030.

To what extent is further developing nuclear energy a sustainable way of generating more electricity in Pakistan? Give reasons to support your judgement and refer to examples you have studied. You should consider different points of view in your answer. [6]

More sustainable because:

- ✓ Boosts economy;
- ✓ Can bridge the gap with energy shortages / deficiencies in oil and gas;
- ✓ Assists development;
- ✓ Provides jobs;
- ✓ A small quantity of uranium can generate a large amount of energy; Less than half kg of uranium contains 3 million more times energy than the same weight of coal;

- ✓ The chances of accidents in nuclear power station is low. There have been fewer accidents in nuclear power stations than any other kind of power station;
- ✓ Nuclear power can help speed up the process of industrialisation;
- ✓ Nuclear power contributes less to the greenhouse effect and acid rain compared to fossil fuels

Less sustainable because:

- ✓ Expensive to build so may have to borrow money or seek investment from other countries / economic burden;
- ✓ Will take up valuable land space needed for more important development projects
- ✓ Renewable energy schemes such as solar energy / wind power are more appropriate;
- ✓ Only provides jobs in the short term whilst building them;
- ✓ Fuel rods in reactors produce dangerous rays which are cancer causing;
- ✓ Nuclear waste remains radioactive for many years;
- ✓ Finding suitable locations for storing radioactive waste is a problem;

Question 4 **REVIEW BOOK BY MYM**

The natural gas field discovered at Sui is considered to be one of the largest in the world. Additional smaller gas fields have since been discovered and developed in Pakistan.

To what extent is further development of the natural gas industry possible in Pakistan? Give reasons to support your judgement. You should consider different points of view in your answer.

Possibilities

- ✓ More pipelines and gas fired thermal power stations could be set up;
- ✓ Potential for more gas fields to be found;
- ✓ Pipeline could be extended further to areas currently not served;
- ✓ Pipeline could be over-ground, doesn't have to be underground;
- ✓ If more gas fired thermal power stations are built, Pakistan could reduce imports of oil and coal;
- ✓ Finding more gas reserves would increase the domestic supply and increase the number of potential years use;

- ✓ Improve the lives of people in remote rural areas / provide employment opportunities;

Difficulties

- ✓ Financial constraints / loans may have to be taken out;
- ✓ Topography and / or climate may hinder or make difficult the building of more pipelines or further exploration;
- ✓ Cost of exploration and / or building thermal power stations will not be value for money / cost effective;
- ✓ Industries would not be built in these areas anyway as remote / not ideal locations for building further industry;